

NARENDRANATH EDARA

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Data Platforms & Applied AI | Python, SQL, Azure, Fabric. Builds ingestion and recovery for 400 payroll databases. Cut refresh from 30 minutes to under 8, freed 5,000 engineering hours a year, and built an AI context harness the team uses.

TECHNICAL SKILLS

Data Engineering: Python, SQL, ETL, data modeling, data quality, CDC, T-SQL, SQL Server, Microsoft Fabric, Power BI
Applied AI & ML: Claude Code, GitHub Copilot, LLM fine-tuning (QLoRA), RAG, pgvector, NLP, PyTorch, MLflow
Distributed Data: Apache Spark, Structured Streaming, Kafka, Delta Lake, Delta tables, lakehouse architecture
Cloud & DevOps: Azure, Azure DevOps, CI/CD, AKS, Docker, SQL Agent, SSIS, Always On Availability Groups

EXPERIENCE

Data Engineer | *ExponentHR*

July 2024 - Present | Dallas, TX

HR and payroll software provider: owns CDC ingestion, recovery, and releases for 400 SQL Server databases, about 70 TB.

- Cut payroll data refresh from 30 minutes to **under 8 minutes** with idempotent change data capture (CDC) MERGE upserts that became the platform standard, replacing full-table reloads; compute cost fell 67% and failed loads rerun cleanly.
- Freed **5,000 engineering hours** a year by automating the 20-a-day database copy and restore on Contained Always On Availability Groups and making it safe to retry; fixed orphaned CDC jobs, replication races, and stuck restores.
- Built an **AI context harness** for Claude Code that the team now uses: primary/foreign keys and table grain generated from schema metadata plus hand-written business meaning, so AI-assisted SQL joins at the right grain and fixes ship faster.
- Reduced client support tickets **40%** and report queries from 12 seconds to under 4 by moving SQL Server reporting to Microsoft Fabric semantic models with natural-language-to-SQL agent access, plus index and stored-procedure tuning.
- Made failures diagnosable instead of retried blindly: an SSISDB-backed procedure logs which SSIS packages ran, when, and their execution details for failed SQL Agent jobs, and control-table validation flags silent data drift.
- Held CDC recovery after Always On failover **under 1 hour**: found and closed a silent post-failover capture-job failure, then wrote runbooks that turned a specialist-only restore into an on-call procedure.
- Compressed releases from 3 months to **14 days** by owning Azure DevOps CI/CD end to end and aligning dev, test, IT, and business through Scrum; automated SSRS and Power BI releases and gated Copilot-assisted code behind review and tests.
- Saved 15+ engineering hours per sprint by automating delivery reporting in Python over Git and Azure DevOps APIs; wrote the GitHub Codespaces, branch-strategy, and work-item docs new hires use instead of asking.

Data Engineer | *Missouri S&T*

Aug. 2023 - July 2024 | Rolla, MO

University research and cloud infrastructure: built NLP sentiment analysis and Azure AI anomaly detection pipelines.

- Caught a production memory leak 4 hours before an outage with Azure AI Anomaly Detector pipelines that chose an algorithm per time-series profile, reaching more than 95% accuracy across research workloads.
- Coauthored a **peer-reviewed NLP** study comparing VADER and RoBERTa sentiment in Python on TripAdvisor and Yelp museum reviews; tested sentiment against visitor ratings and showed how source and method change the conclusions.
- Cut Azure spend \$3,200 a month by moving anomaly detection from 20 static virtual machines to Azure Kubernetes Service with horizontal pod autoscaling, lifting CPU utilization from 12% to 64% on 4 to 8 nodes.

Product Intern | *C2FO*

June 2023 - Aug. 2023 | Kansas City, KS

B2B fintech working-capital transaction platform: summer product internship working with product and operations.

- Analyzed transaction and user behavior in SQL across a B2B working-capital platform, turning findings into product decisions.
- Halved resource-allocation time by writing preferred-offers product requirement documents from stakeholder interviews.

Business Intelligence Analyst | *Udaan*

Sept. 2020 - Mar. 2021 | Nashik, India

New-city B2B staples marketplace: owned gross merchandise value, demand planning, and procurement for the Nashik launch.

- Grew Nashik gross merchandise value to INR 5 crore a month during the new-city Staples launch, matching the incumbent; reached INR 3.9 crore in December, three months after launch, at 1.3x month-over-month growth.
- Hired and managed a 10+ person sourcing team and sustained 99.3% order fulfillment by applying statistical demand analysis to capacity planning under just-in-time and warehouse models for the new category.
- Kept delivery cycles running through application downtime by devising and executing a business continuity plan.

Business Analyst | Zomato

Mar. 2018 - Sept. 2020 | Hyderabad, India

Food-delivery marketplace across South India: owned pricing intelligence, search analysis, and a 300-restaurant portfolio.

- Turned **contribution per order** from negative 18 to positive 2 rupees across 300 restaurants by rebalancing discount sharing against advertising revenue, moving the portfolio from loss to positive unit economics.
- Contributed to a 9% market-share gain in contested metros by building competitor analytics on price, delivery time, and coverage, with ingestion and refresh that kept pricing on current rather than week-old signals.
- Cut support-desk volume 80% by indexing 100,000+ internal documents in Elasticsearch for self-service search.
- Opened operations in 4 new cities on the Project C100 launch team, reaching 1,000 orders a day within 3 days in Guntur.

PROJECTS

repo-context-hooks | Python, GitHub Actions, Sigstore, CodeQL

PyPI

- Published a zero-dependency Python package that snapshots repository context around each commit so LLM coding agents resume work across sessions, with Sigstore-signed releases, CodeQL on every pull request, and **330+ passing tests**.

FinTune | Mistral-7B, QLoRA, FastAPI, PyTorch

2026

- Fine-tuned Mistral-7B with QLoRA (4-bit NF4) for financial sentiment and served it through FastAPI with PII redaction before inference and KL-divergence drift monitoring; 35+ tests cover data, model, and serving.
- Added a three-state circuit breaker whose recovery actions cut batch size on out-of-memory errors, fall back to the quantized model on latency spikes, and reload with exponential backoff after load failures.

Real-Time Fraud Detection Platform | Kafka, FastAPI, LightGBM, MLflow, Docker

2026

- Built Kafka-to-FastAPI fraud scoring on 100,000 synthetic transactions with an MLflow-tracked LightGBM model, idempotent Postgres writes so replays are no-ops, and Prometheus and Grafana monitoring; P99 scoring latency 1.12 ms per request.

AutoApply AI | Chrome MV3, FastAPI, pgvector, React

2026

- Built a privately deployed Chrome extension and FastAPI backend that grounds application answers in prior work history with pgvector RAG and routes across six LLM providers; 11 ATS adapters, 40+ endpoints, **355 backend tests**.

Portfolio Risk Analytics (in progress) | Kafka, Spark Structured Streaming, Delta Lake, FastAPI

2026

- Built Kafka-to-Spark Structured Streaming ingestion of market ticks into a Delta Lake bronze layer with checkpointing, 5-second windows, and a 10-second watermark; the FastAPI value-at-risk handoff is still in progress.

EDUCATION & CERTIFICATIONS

Missouri S&T - MS Information Science and Technology, GPA 4.0/4.0

Jan. 2022 - Dec. 2023

GVP College of Engineering - B.Tech Mechanical Engineering

2017

DP-700 Microsoft Certified: Fabric Data Engineer Associate | Certified Scrum Product Owner (CSPO)

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